

Chun-Wei Chen

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Education

National Tsing Hua University (NTHU)

Hsinchu, Taiwan

- College of Semiconductor Research
- Advisor: Prof. Ting-Chi Wang

2025/09 - Present

National Central University (NCU)

Taoyuan, Taiwan

- Bachelor of Science in Electrical Engineering
- BS GPA: 4.0/4.3

2021/09 - 2025/06

Experience

NSTC Project - Thermal-Aware Global Routing for SoC Reliability

2025/09 - Present

Advisor: Prof. Ting-Chi Wang

- Developing a thermal-aware electrical global router for SoC physical design by incorporating hotspot distribution into multi-objective routing cost optimization, including overflow, wirelength, via count, and thermal hotspot avoidance.
- Building a PyTorch-based ML thermal map predictor using a U-Net with attention architecture, with planned training data generated from thermal simulation tools such as Ansys Icepak, Cadence Celsius, and HotSpot, to support high-speed in-loop thermal inference during routing.
- Investigating extensions toward optical-electrical co-design by considering routing resource allocation, hotspot distribution, and the thermal sensitivity of optical devices.

2024 ICCAD CAD Contest: Problem D - Chip Level Global Router

2024/06 - 2024/09

Advisor: Prof. Yu-Guang Chen

- Recognized with an **Honorable Mention**.
- Developed methods to optimize four key aspects of global routing at the chip level: wirelength, channel overflow, via count, and edge pin density.
- Implemented an Obstacle-Avoiding Rectilinear Spanning Tree (OARST) algorithm to minimize routing length while navigating around obstacles
- Applied a modified A-star algorithm to further reduce via count and edge pin density after the OARST stage.

Python-Based SPICE Circuit Simulation

2023/06 - 2024/05

Advisor: Prof. Chiu-Chih Chou

- Developed a Python-based SPICE circuit simulator supporting AC, DC, and transient (TRAN) analyses.
- Implemented a SPICE netlist parser capable of handling subcircuit statements.
- Applied four numerical methods—Newton's Method, Backward Euler, 2-step Backward Differentiation Formula (BDF2), and Modified Nodal Analysis (MNA)—to ensure accurate circuit simulation.
- Successfully simulated circuits with up to 234 MOSFETs, achieving an average error of only 1–2% compared to Ngspice.

Skills

- *Programming Languages: C/C++, Python, Git.*
- *English ability: TOEIC 895 (Golden certificate)*
- *Tools: Cadence Voltus IC Power Integrity Solution, Cadence Innovus, Cadence Virtuoso, Cadence Allegro Package Designer, Synopsys HSPICE, OpenROAD-flow-script.*
- *Courses: VLSI PDA (Physical Design Automation), Data Structure, Algorithms, Machine Learning for EDA, Digital System Design and Implementation, Computer Architecture, Operating System, Essentials of Intelligent System Design Flow, Semiconductor Materials.*

Academic Project

Global Router (C++) | VLSI Physical Design Automation

Fall 2025

- *Implemented a C++ grid-based global router on a DRC-clean baseline, modeling GCell edges with capacity, demand, and history-based costs.*
- *Applied congestion-aware A* maze routing to reduce overflow while keeping wirelength competitive with the baseline solution.*

Detailed Placement (C++) | VLSI Physical Design Automation

Fall 2025

- *Built a C++ standard-cell detailed placement optimizer with Global Swap, Vertical Swap, Local Reordering, and Single-Segment Clustering.*
- *Designed incremental cost with partial HPWL and per-net min/max cache, supporting move-to-empty and mutual-OR checks; candidates retrieved via bin/segment index.*

k-way Min-cut Partitioning (C++) | VLSI Physical Design Automation

Fall 2025

- *Implemented a high-performance Fiduccia–Mattheyses (FM) k-way hypergraph partitioner in C++ for large-scale netlists.*
- *Designed custom bucket-list data structures and Δ -gain update mechanisms to support balanced 2-way and recursive 4-way partitioning.*
- *Optimized runtime and memory efficiency through intrusive linked lists, gain caching, and incremental updates.*